BankWise AI

Smart Banking Customer Service & Advisory Agent

Project Documentation

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Project Team

Ahmed Nasser Abdelaziz Youssef (Team Leader)

Mohamed Ashraf Farghaly Mohamed

Noran Mounir Mohamed El-Awady

Abdelrahman Yasser Refaat Badawy

Basma Ahmed Mahmoud Bayoumi

Mohamed Ashraf Salem Ahmed

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1. Project Proposal

1.1 Project Overview

BankWise AI is a proposed AI-powered assistant designed to make common banking information easier to access and customer-service guidance more efficient. Through a chat interface, users can ask banking-related questions, receive answers based on an available knowledge base, get help categorizing complaints, and obtain preliminary illustrative loan or credit eligibility estimates.

The project combines a conversational AI agent with Retrieval-Augmented Generation (RAG), session context, basic request-planning capabilities, and a monitoring dashboard. It aims to demonstrate how artificial intelligence can support routine banking inquiries while providing traceable answers and handling uncertainty responsibly.

Chat & Q&A	Complaints	Eligibility	Dashboard
Banking questions and follow-ups with source references	Classification and department routing suggestions	Illustrative loan / credit estimates	FAQs, usage, complaint categories, token cost

1.2 Project Objectives

The main objectives of BankWise AI are to:

- Provide a simple chat interface for common banking questions and follow-up queries.
- Retrieve relevant information from banking FAQs and representative documents and provide source references where available.
- Classify customer complaints into predefined categories and suggest the appropriate department for routing.
- Offer preliminary loan or credit eligibility estimates using transparent, predefined illustrative rules.
- Maintain relevant conversation context during a session and acknowledge when available information is insufficient to provide a reliable answer.
- Provide an internal dashboard for monitoring frequently asked questions, complaint categories, usage statistics, and available token usage and cost estimates.

1.3 Project Scope

1.3.1 In Scope

The project will cover the following components:

Component	Description
Agent System	A conversational assistant that answers banking-related questions, maintains session context, classifies complaints, suggests routing destinations, and calculates preliminary loan or credit eligibility estimates using predefined illustrative rules.

Component	Description
RAG Integration	A searchable knowledge base containing representative banking FAQs and documents. The assistant will retrieve relevant information, provide source references where available, and communicate uncertainty when the available evidence is insufficient.
User Interface and Dashboard	A customer chat interface and an internal monitoring dashboard for reviewing frequently asked questions, complaint categories and routing suggestions, usage statistics, and token usage or estimated conversation costs when supported by the selected model provider.
Deployment and Security	Basic authentication and access control for internal features, secure handling of API keys and other secrets, error handling, and rate limiting where applicable.

1.3.2 Data Sources and Technical Approach

The team will initially use synthetic data and clearly labeled illustrative sample documents. Publicly available official banking information may be added when suitable sources are identified. Any team-created examples of banking policies, fees, or eligibility rules will be labeled illustrative and will not be presented as actual bank policies.

Candidate technologies include Python, a lightweight application framework such as Streamlit, a large language model (LLM), embedding models, and a vector database such as FAISS or ChromaDB. Final technology choices will be made during implementation planning.

1.3.3 Out of Scope and Project Boundaries

The prototype will not execute real banking transactions, access live customer accounts, or make official lending decisions. It will not connect to private banking systems unless an explicitly authorized integration becomes available. Loan or credit eligibility results will be preliminary illustrative estimates only. Complaint routing will recommend an appropriate destination but will not submit complaints to a real bank system.

2. Literature Review

2.1 Purpose and Scope

This review surveys existing banking assistants, the academic and technical work behind them, and the risks that regulated financial services add to conversational AI. Its goals are to identify what comparable projects do well and where they fall short, to position BankWise AI against them, and to justify the design decisions made in this project. The review covers four areas that mirror the four capstone pillars: conversational agents and tool use, retrieval-augmented generation (RAG) over policy documents, Arabic and English language support, and security, privacy and monitoring in a banking context.

2.2 Similar Projects and Solutions

2.2.1 Commercial Banking Assistants

Large banks worldwide have deployed virtual assistants for several years. The table below summarizes the best-known examples and the typical Egyptian market baseline. Descriptions are based on publicly available information from each organization.

System	What it does	Strengths	Limitations
Erica Bank of America (USA)	In-app voice and text assistant for balances, transactions, bill reminders, spending insights and proactive alerts.	Very large user base; proactive insights; tightly integrated with account data; long production track record.	English and Spanish only; closed and proprietary; mainly transactional, with limited explanation of policy or fee documents; no Arabic.
Eno Capital One (USA)	SMS and app assistant for balances, payments, fraud alerts and purchase notifications.	Lightweight channel (SMS); strong fraud and alert use cases; low friction.	Narrow, intent-based scope; limited multi-step reasoning; English only.
Ceba Commonwealth Bank (Australia)	In-app assistant for account queries, payments and card support.	Good integration with the bank's mobile app; handles common service tasks.	Tied to one bank's app; limited advisory or eligibility reasoning; no Arabic or dialect support.
Egyptian bank chatbots (typical baseline)	Menu- or keyword-driven bots in mobile apps and social messaging channels, often with hand-off to a call center.	Familiar channels; reduce some call-center load; embedded in existing apps.	Mostly scripted FAQs; weak handling of Egyptian Arabic and mixed Arabic/English; limited memory; little transparency on how answers are produced.

The Egyptian row describes the general pattern of publicly visible service channels. Details differ from one bank to another.

2.2.2 Open-Source and Research Approaches

Classic task-oriented frameworks such as Rasa (Bocklisch et al., 2017) combine intent classification, entity extraction and dialogue management. They give predictable, auditable behavior, which suits banking, but they require substantial labelled data and manual dialogue design, and they struggle with questions outside the trained intents. The BANKING77 dataset (Casanueva et al., 2020) provides 77 fine-grained banking intents and shows that pre-trained sentence encoders can reach strong intent-detection accuracy with limited data. This is directly relevant to the complaint-category classifier, although the dataset is English-only and does not reflect Egyptian products or dialect.

Large language models (LLMs) removed much of the manual design effort but introduced hallucination, where a fluent answer is not grounded in fact (Ji et al., 2023). Retrieval-augmented generation (Lewis et

al., 2020) addresses this by retrieving relevant documents at query time and conditioning the answer on them. Gao et al. (2023) survey RAG designs and highlight chunking, retrieval quality and evaluation as the main levers. Liu et al. (2024) show that models use information placed in the middle of long contexts less reliably, which supports retrieving a small number of precise chunks rather than many long ones. Es et al. (2023) propose RAGAS, a reference-free framework for measuring faithfulness and answer relevance, which informs the evaluation plan.

2.3 Technical Foundations Relevant to BankWise AI

Tool-using and multi-step agents. ReAct (Yao et al., 2023) interleaves reasoning with actions, and Toolformer (Schick et al., 2023) shows that language models can learn when to call external tools. These ideas underpin the design of BankWise AI: the loan-eligibility calculator and the complaint-routing classifier are deterministic tools that the agent calls, rather than numbers the language model invents. Compound requests, such as a loan question plus an address update, are handled by planning the steps first and executing them in sequence.

Domain-specific language models. FinBERT (Araci, 2019) and BloombergGPT (Wu et al., 2023) show that financial-domain adaptation helps on financial text. However, both focus on English financial news and analysis rather than customer service, and training such models is far beyond a capstone budget. The project therefore uses a general LLM grounded by RAG over bank documents.

Arabic and bilingual NLP. AraBERT (Antoun et al., 2020), ARBERT and MARBERT (Abdul-Mageed et al., 2021) and CAMeLBERT (Inoue et al., 2021) show that Arabic-specific models outperform multilingual ones on many tasks, and MARBERT in particular targets dialectal Arabic including Egyptian. For retrieval, multilingual embedding models such as multilingual-E5 (Wang et al., 2024) allow an Arabic question to retrieve an English policy passage and vice versa. This matters because Egyptian customers often mix Arabic and English, and bank documents may exist in either language.

Chunking formal policy text. Terms and conditions and fee schedules contain numbered clauses, tables and cross-references. Fixed-size splitting can separate a fee from its condition. Surveys of RAG practice (Gao et al., 2023) recommend structure-aware chunking with overlap and metadata. The project applies this by splitting on clause and heading boundaries and keeping the document name and section in each chunk's metadata so that answers can cite their source.

2.4 Risk, Security and Compliance Literature

Banking is a high-stakes domain, so the literature on safe deployment matters as much as model quality. Three themes recur.

- **Wrong answers have legal and financial consequences.** In Moffatt v. Air Canada (2024 BCCRT 149), a Canadian tribunal held the airline responsible for incorrect information given by its website chatbot. The lesson for BankWise AI is to answer only from retrieved policy text, show the source, and escalate to a human when evidence is missing.
- **LLM-specific attack surface.** The OWASP Top 10 for LLM Applications lists prompt injection, sensitive information disclosure and excessive agency as leading risks. The planned mitigations are input validation, least-privilege tools, no account data in prompts beyond the authenticated session, and output checks.
- **Governance and privacy.** The NIST AI Risk Management Framework (2023) recommends mapping, measuring and managing AI risks throughout the lifecycle. In Egypt, the Personal Data Protection Law No. 151 of 2020 regulates the handling of personal data, and the Central Bank of

Egypt promotes digital banking and financial inclusion. These motivate the authentication, access control, secrets management, rate limiting and logging design.

2.5 Comparison with BankWise AI

Criterion	Erica / Eno / Ceba	Typical Egyptian bank bots	Rasa-style frameworks	BankWise AI
Arabic + English	No Arabic	Limited, often scripted	Possible but needs data	Both languages, mixed input supported
Answers from policy documents	Limited	Static FAQs	Needs manual intents	RAG over policies, fees and FAQs with source citation
Calculation tools	Account-specific	Rare	Custom actions	Loan eligibility calculator as a tool
Complaint routing	Basic hand-off	Hand-off to call center	Intent classifier	Category classifier with ambiguity handling
Multi-step requests	Limited	No	Via stories and rules	Agent planning with session memory
Staff monitoring dashboard	Internal, not public	Not visible	Not included	Usage, top questions, escalation rate, token cost
Transparency and auditability	Proprietary	Low	High	Open design, logs and documented limits

2.6 Gap Analysis and Lessons Applied

The literature reveals a gap that this project targets. Global assistants are mature but proprietary, English-centered and focused on transactions. Local assistants cover the right languages and channels but tend to rely on scripted flows. Research shows that RAG and tool use can reduce hallucination and support multi-step tasks, but few public projects combine them with Egyptian-market requirements: bilingual and dialect-aware dialogue, bank-policy grounding, and production-grade controls. BankWise AI addresses this by combining the following elements.

- **Grounding over generation:** answers come from retrieved policy chunks, with citations, to limit hallucination (Lewis et al., 2020; Ji et al., 2023).
- **Deterministic tools for numbers and routing:** eligibility and complaint categories are computed by tools, not guessed (Yao et al., 2023; Schick et al., 2023).
- **Bilingual retrieval:** multilingual embeddings let Arabic and English queries reach the same knowledge base (Wang et al., 2024).
- **Safety by design:** authentication, secrets management, rate limiting and structured error handling follow OWASP and NIST guidance.
- **Measured quality:** a 10-question benchmark, latency targets and faithfulness checks inspired by RAGAS (Es et al., 2023).

Limitations of existing work that also apply here. Most evaluations use English benchmarks, and Egyptian dialect resources are scarce, so bilingual accuracy has to be tested on the project's own sample questions. Public bank policy documents are also limited, so representative documents are used instead of live bank data. These points are stated openly in the limitations write-up.

3. Requirements Gathering

3.1 Stakeholder Analysis

3.1.1 Introduction

Stakeholder analysis identifies the main groups that interact with BankWise AI or are affected by its operation. It helps clarify their expected needs and ensures that the proposed system addresses its intended users, operational objectives, and project limitations.

The stakeholders and needs identified in this section are based on the official project brief and the proposed project scope. They represent preliminary requirements that may be reviewed and refined during system analysis and design.

3.1.2 Key Stakeholders and Their Needs

Banking Customers

Role: Primary users who interact with the assistant to obtain banking information and guidance.

Needs:

- Receive clear answers about banking services, account-related inquiries, fees, commissions, and loan eligibility requirements.
- Interact with the assistant in Arabic or English.
- Ask follow-up questions while retaining relevant conversation context.
- Receive source references when answers are based on retrieved documents.
- Be informed when sufficient evidence is unavailable to provide a reliable answer.
- Receive preliminary loan eligibility estimates based on the documented rules available in the prototype.

Customer Support Staff

Role: Intended users of the internal monitoring dashboard, as specified in the project brief.

Needs:

- Review frequently asked questions and complaint categories.
- Monitor usage and escalation indicators.
- Review suggested complaint-routing destinations.
- Access available monitoring information through an appropriate internal interface.
- Use internal dashboard features protected by authentication and access controls.

The dashboard is part of the proposed prototype. Its final implementation details and operational use will be confirmed during system analysis and design.

Project Team and Developers

Role: Design, implement, test, and maintain the BankWise AI prototype.

Needs:

- Develop a modular and maintainable system architecture.

- Manage and update the representative banking knowledge base.
- Evaluate answer quality, complaint classification, and response latency.
- Monitor token usage and estimated model costs where supported by the selected provider.
- Protect API credentials and other secrets.
- Handle errors securely and provide appropriate fallback messages.
- Document limitations and evaluate system performance against the agreed success criteria.

Banking Domain Experts

Role: Potential reviewers who may validate banking terminology, representative policies, eligibility rules, and expected answers if access to such expertise becomes available.

Needs:

- Review the accuracy and clarity of banking information.
- Identify incorrect, ambiguous, or potentially misleading answers.
- Verify that illustrative sample documents and eligibility rules are not presented as official bank policies.

Participation by banking domain experts has not yet been confirmed.

Project Evaluators

Role: Assess the project against its documentation, implementation, and evaluation criteria.

Needs:

- Review clearly documented requirements and project boundaries.
- Verify system behavior through representative test cases.
- Review evidence of answer quality, complaint classification, and response latency.
- Assess the implementation of authentication, access controls, and secure secret handling.
- Review documented limitations and proposed solutions for future production use.

3.1.3 Stakeholder Analysis Summary

Banking customers are the primary end users of BankWise AI, while customer support staff are the intended users of the internal monitoring dashboard. Developers are responsible for building and maintaining the prototype. Banking domain experts may support content validation if available, and project evaluators assess whether the documented objectives and requirements have been met.

The identified needs are preliminary. They are derived from the project brief rather than from completed interviews with actual customers or banking staff.

3.2 User Stories and Use Cases

3.2.1 Introduction

User stories and use cases describe how intended users interact with BankWise AI to accomplish common tasks. User stories express user needs and expected benefits, while use cases describe the steps involved in a particular interaction, including alternative scenarios where appropriate.

The scenarios below are derived from the proposed project scope and represent intended prototype behavior. They do not imply that the features have already been implemented or tested.

3.2.2 User Stories

US-01: Ask a Banking Question

User Story: As a banking customer, I want to ask questions about banking services, fees, and requirements so that I can understand relevant banking information more easily.

Acceptance Criteria:

- The user can submit a question through the chat interface.
- The system provides an answer based on the available information.
- The system communicates when reliable supporting information is unavailable.

US-02: Receive Document-Based Answers

User Story: As a banking customer, I want answers to be supported by relevant banking documents so that I can understand where the information comes from.

Acceptance Criteria:

- The system retrieves relevant information from the available knowledge base.
- Available source references are displayed with the answer.
- Unsupported information is not presented as verified fact.

US-03: Check Preliminary Loan Eligibility

User Story: As a banking customer, I want to provide the information required by the loan eligibility calculator so that I can receive a preliminary estimate based on the prototype's documented eligibility rules.

Acceptance Criteria:

- The system requests the inputs required by the configured rules.
- The system evaluates the provided information against those rules.
- The result is clearly labeled as an illustrative estimate.
- The system does not present the result as an official bank approval or rejection.

US-04: Classify a Complaint

User Story: As a banking customer, I want to describe my complaint so that the system can identify its likely category and suggest an appropriate support destination.

Acceptance Criteria:

- The system assigns a category from the predefined complaint categories.
- A suggested routing destination is displayed.
- The system communicates uncertainty when classification is unclear.
- The system does not claim that the complaint has been submitted to a real bank.

US-05: Continue a Conversation

User Story: As a banking customer, I want the assistant to retain relevant context during an active session so that I can ask follow-up questions without repeating all previous details.

Acceptance Criteria:

- The system uses relevant information from the active conversation.
- Follow-up responses remain consistent with the available context.
- Session context is handled according to the prototype's privacy requirements.

US-06: Monitor System Usage

User Story: As a customer support staff member, I want to review usage statistics, frequently asked questions, complaint categories, and available cost indicators so that I can understand how the prototype is being used.

Acceptance Criteria:

- The internal dashboard displays the supported monitoring indicators.
- Internal features are protected by appropriate authentication and access controls.
- Token usage and estimated costs are displayed when supported by the selected provider and implementation.

3.2.3 Use Cases

UC-01: Answer a Banking Question

Primary Actor: Banking Customer

Preconditions:

- The chat interface is available.
- The knowledge base contains relevant documents, where applicable.

Main Flow:

1. The customer enters a banking question.
2. The system processes the question.
3. The retrieval component searches for relevant information in the knowledge base.
4. The system generates an answer based on the retrieved evidence.
5. The system displays the answer and available source references.

Alternative Flow:

- If sufficient evidence cannot be found, the system informs the customer that it cannot provide a reliable answer and recommends checking an official banking channel when appropriate.

Postcondition: The customer receives a supported answer or a clear message explaining that sufficient evidence is unavailable.

UC-02: Estimate Loan Eligibility

Primary Actor: Banking Customer

Preconditions:

- Illustrative eligibility rules have been documented and configured in the prototype.

Main Flow:

1. The customer selects the preliminary loan eligibility feature.
2. The system requests the information required by the configured rules.
3. The customer provides the requested information.
4. The system evaluates the inputs against the documented eligibility rules.
5. The system displays the preliminary result and explains its illustrative nature.

Alternative Flow:

- If required information is missing or invalid, the system requests correction or additional information.

Postcondition: A preliminary rule-based eligibility estimate is displayed. No official loan decision is made.

UC-03: Classify and Route a Complaint

Primary Actor: Banking Customer

Preconditions:

- The predefined complaint categories and routing suggestions are available.

Main Flow:

1. The customer enters a complaint description.
2. The system analyzes the text.
3. The system assigns the most suitable available complaint category.
4. The system identifies a suggested support destination based on the category.
5. The system displays the classification and routing suggestion.

Alternative Flow:

- If the complaint is ambiguous or does not match the available categories, the system requests clarification or indicates that the classification may be uncertain.

Postcondition: A suggested complaint category and destination are displayed. The complaint is not submitted to a real bank system.

UC-04: Monitor System Activity

Primary Actor: Authorized Customer Support Staff Member

Preconditions:

- The internal dashboard has been implemented.
- The user has the required authorization.
- Relevant monitoring data is available.

Main Flow:

1. The staff member opens the internal dashboard.
2. The system verifies the user's access permissions.

3. The dashboard displays available usage statistics, frequently asked questions, complaint categories, and supported token or cost indicators.
4. The staff member reviews the available information.

Alternative Flow:

- If authentication fails, the system denies access to restricted dashboard information.
- If monitoring data is unavailable, the dashboard displays an appropriate message.

Postcondition: The authorized user can review the available monitoring information.

3.2.4 Assumptions and Limitations

- The user stories and use cases describe intended behavior and must be reviewed against the final prototype.
- The loan eligibility calculator will provide preliminary estimates based on documented illustrative rules, not official bank decisions.
- Complaint classification and routing are suggestions. Actual complaint submission to a bank system is outside the current project scope.
- The dashboard is intended for customer support monitoring, but its final implementation details and operational use remain subject to system analysis and design.
- The prototype will not perform real banking transactions or access live customer accounts or confidential banking databases.

3.3 Functional Requirements

Functional requirements specify the features and actions that BankWise AI is expected to provide.

ID	Category	Requirement
FR-01	Chat Interface	The system shall provide a conversational interface through which users can submit banking questions and receive responses in the supported languages.
FR-02	Knowledge-Based Answers	The system shall answer supported banking questions using relevant information retrieved from the available knowledge base.
FR-03	Source References	The system shall display available source references for answers generated using retrieved documents.
FR-04	Complaint Classification	The system shall classify customer complaints into predefined categories.
FR-05	Complaint Routing Suggestions	The system shall suggest an appropriate support department or destination based on the predicted complaint category. It shall not claim to submit complaints to a real bank system.
FR-06	Preliminary Loan Eligibility Estimation	The system shall provide a preliminary loan or credit eligibility estimate based on documented illustrative rules and user-provided information.
FR-07	Session Context	The system shall retain and use relevant context during an active conversation to support follow-up questions.
FR-08	Insufficient Evidence Handling	The system shall inform users when the available evidence is insufficient to provide a reliable answer.
FR-09	Internal Monitoring Dashboard	The system shall provide an internal dashboard intended for customer support monitoring, displaying supported usage statistics, frequently asked questions, and complaint or escalation indicators.

ID	Category	Requirement
FR-10	Token and Cost Display	The system shall display token usage and estimated model costs per conversation when the selected provider and implementation make these metrics available.

3.4 Non-Functional Requirements

Non-functional requirements specify the expected quality, performance, security, usability, privacy, and maintainability of the system.

ID	Category	Requirement
NFR-01	Performance	The system should respond within 10 seconds for at least 90% of test queries under normal operating conditions. This is a proposed target that must be confirmed after selecting the model and deployment environment.
NFR-02	Answer Quality	The system shall correctly answer at least 80% of a predefined set of representative sample banking questions, measured against an agreed answer key or evaluation procedure.
NFR-03	Secret Protection	The implementation shall store credentials and API keys outside committed source code and shall not expose them through application output, logs, or error messages.
NFR-04	Access Control	The internal dashboard shall require authentication, and restricted features shall be accessible only to authorized users.
NFR-05	Error Handling	The system shall handle errors gracefully and provide clear fallback messages without exposing stack traces, credentials, or other sensitive implementation details.
NFR-06	Usability	The interface should allow a first-time user to submit a question and understand the response without requiring technical instructions.
NFR-07	Maintainability	The application should use modular components so that the interface, retrieval pipeline, agent workflows, and dashboard can be updated and tested independently where practical.
NFR-08	Privacy	Testing shall use synthetic, public, or appropriately authorized data. Real account credentials and unnecessary sensitive customer information shall not be used.
NFR-09	Extensibility	The system should allow developers to add supported documents and complaint categories without requiring a complete rewrite of the application.
NFR-10	Responsible AI	The system shall label illustrative eligibility results clearly, avoid claims of official loan approval or rejection, communicate uncertainty, and direct users to official banking channels when confirmation is needed.

3.5 Validation Notes

- The 80% answer-quality target follows the project's stated success criterion. It should be measured using a predefined set of sample questions and an agreed answer key.

- The 10-second response-time target is a proposed working target. It should be confirmed after selecting the model and deployment environment.
- Security, access control, and privacy requirements must be verified through implementation and testing.
- The requirements should be reviewed against the final prototype and documented test results before project submission.

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